

CapsQual: GUI-Tool for Transforming Subtitles into Qualitative Research Transcripts

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Software

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Summary

CapsQual is a free open-source GUI-workstation for transforming subtitle files created by automatic speech recognition (ASR) software such as Whisper ([OpenAI, 2023](#)) or CapsWriter-Offline ([HaujetZhao, 2023](#)) into qualitative transcripts following established transcription conventions such as GAT2 ([Selting et al., 2011](#)), TiQ ([Przyborski & Wohlrab-Sahr, 2021](#)), Kuckartz & Rädiker ([2024](#)) and Dresing & Pehl ([2019](#)). Users can quickly assign speakers to segments, enter transcription symbols and sequentially edit and refine transcripts, thus aiding in their preparation for analysis within or without computer-assisted qualitative data analysis software (CAQDAS). The tool facilitates the automatic implementation of basic formatting templates, while offering ample options for customization.

Statement of need

The introduction of ASR-assisted transcription workflows resulted in a substantial reduction of manual labor, since qualitative researchers no longer have to type in utterances key by key ([Wollin-Giering et al., 2024](#)). However, while current trends in transcription seem to be pointing toward more automation and less need for manual intervention, researchers largely agree that manual transcription has several benefits for qualitative research: The process involves interpretative actions as well as data reduction tailored to the research needs and results in more familiarity with the material ([Dörnyei, 2007](#)). Unfortunately, manual transcription remains labor-intensive and professional software for manual transcription such as EXMARaLDA ([Schmidt & Wörner, 2014](#)) has a steep learning curve, while fully automated open-source solutions such as noScribe ([Dröge, 2025](#)) are scarce and still require manual editing in order to achieve the required outcome. Unlike all-in-one solutions which implement ASR-models to achieve a direct transcription of audio material, CapsQual aims to fill the niche function of assisting in the process of sequentially converting ASR output into interview transcripts for different qualitative purposes such as conversation analysis or reconstructive analysis.

State of the field

Though many server-based providers of automated transcription services have appeared over the past few years, very few of them offer customization or multiple transcription options. Some distributors of proprietary CAQDAS, such as MAXQDA ([VERBI Software, 2024](#)), offer cloud-based services to transcribe qualitative data for analysis. NoScribe ([Dröge, 2025](#)) is an open-source program that implements various speech recognition models to produce diarized transcripts following conventions common in qualitative and thematic analysis. NoScribe also includes an editor for manually reviewing transcripts. GailBot ([Umair et al., 2022](#)) is an API-

40 based software-package that combines speech-to-text systems and various plugins to produce
41 Jefferson transcripts. EXMARaLDA (Schmidt & Wörner, 2014) is a linguistics-worksuite aimed
42 at producing accurate transcriptions from video or audio material, arranged in **Partitur**-layouts
43 resembling musical scores. No dedicated tools are known for assisting in the semi-manual
44 annotation or preparation of GAT2 or TiQ transcripts from ASR-output.

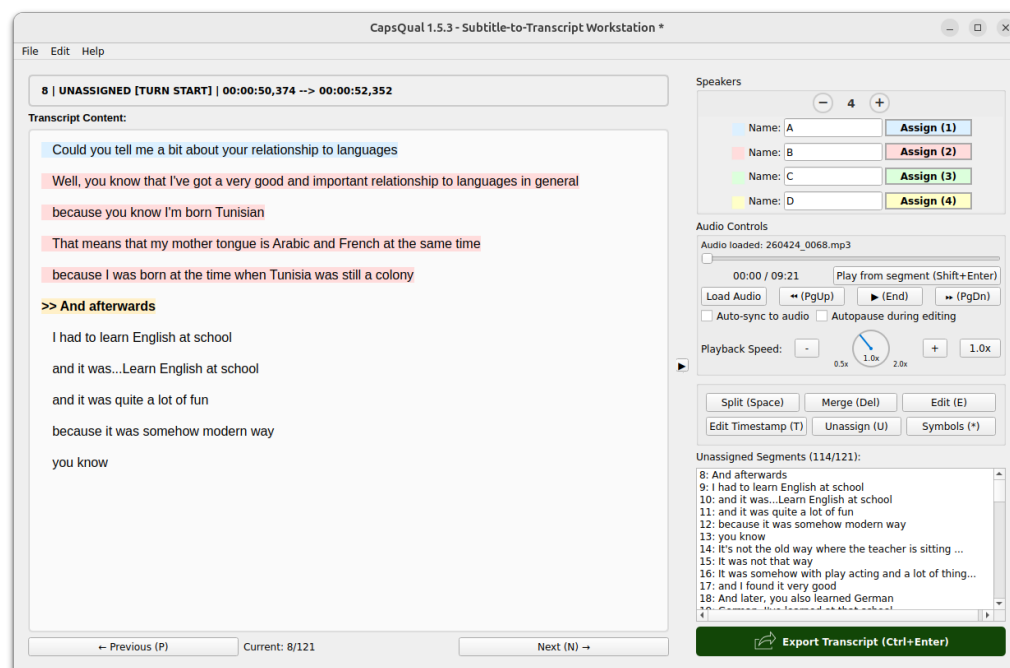


Figure 1: Assigning speakers in CapsQual 1.5.3

45 Software design

46 CapsQual was written in Python, using the PyQt5 framework for cross-platform compatibility
47 (it was tested on Linux, macOS and Windows). The audio player implements python-vlc to
48 realize playback speed adjustment. A fallback player (without speed adjustment) based on
49 Pyaudio is also available. Special attention was given to keyboard-accessibility and compatibility
50 with Chinese, Japanese and Korean (CJK) transcripts. The CapsQual code is split into modules
51 which are subject to continuous integration and unit tests. CapsQual imports subtitle and audio
52 files that can be synced if timestamps are available. Speech-to-text is not integrated, since
53 CapsQual is designed to deal with existing ASR-output as a stand-alone application. The initial
54 transcript content is viewed in segments, which are sequentially assigned to speakers, reviewed
55 and annotated using keyboard shortcuts or UI buttons (see Figure 1). Comments, pauses
56 and other transcription symbols can be chosen from a symbols dialog with tabs for different
57 conventions. Custom symbols can also be configured, saved and loaded. Besides the sequential
58 editing, the program offers automated modification functions: stripping punctuation, converting
59 to lowercase and calculating pauses for segments without voice activity. The export function
60 reshapes the transcript content based on one of three customizable transcription conventions
61 commonly used for thematic, reconstructive and conversation analysis (see Figure 2).

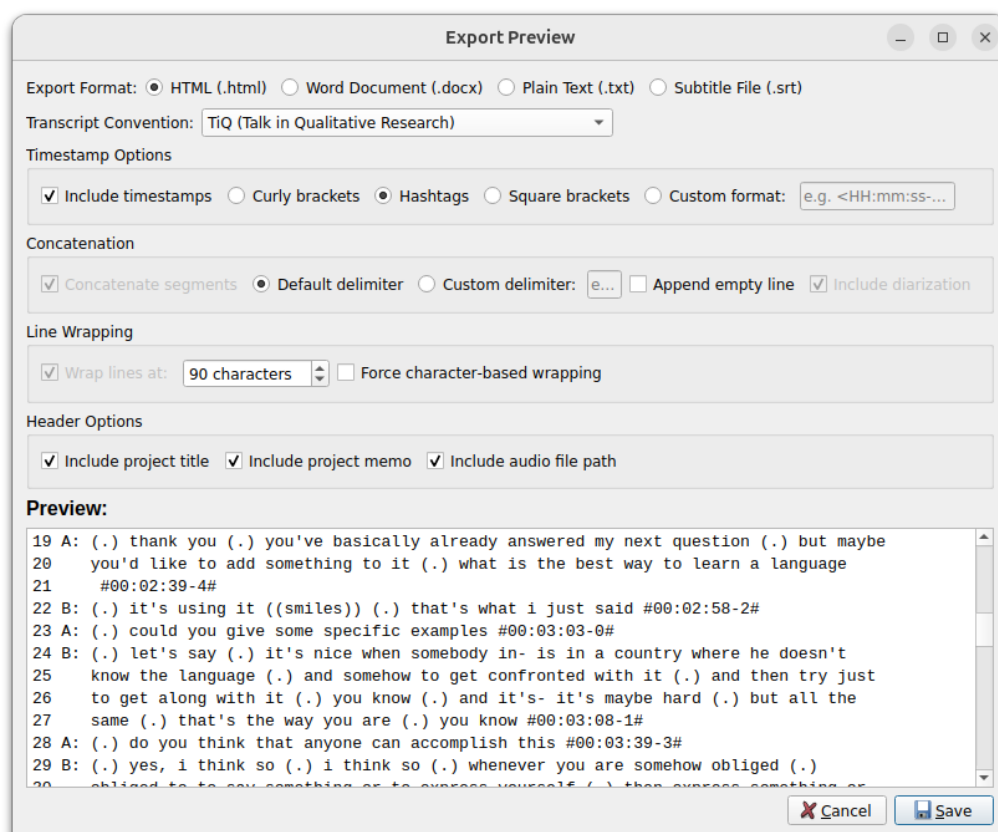


Figure 2: Exporting a transcript in CapsQual 1.5.3

Research impact statement

CapsQual was written for a very specific purpose: to bridge the gap between openly available ASR-models and CAQDAS. Personal research needs led to its development - it is a growing project that has been used for studies related to L2-motivation in multilingual contexts (e.g. Gadermann, in print). The workflow proposed by CapsQual represents a compromise between the convenience of AI-transcription and data-familiarity associated with manual transcription. The open-source tool provides a freely available and user-friendly means of preparing qualitative data for analysis efficiently.

AI usage disclosure

CapsQual was developed using DeepSeek (Version 3.2). The large language model chat interface was used to draft the initial Python scripts. As the project closed in on the desired shape, AI usage turned more towards fine-tuning, testing and debugging of the interface, audio playback, export and other functions. Besides DeepSeek, Cursor (Version 3.1.17) was used on few occasions for refactoring and debugging.

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